EXTENSION OF SCALARS PRESERVES FLATNESS: \MATHCAL{G} FLAT OVER \MATHCAL{O}_1 \RIGHTARROW \MATHCAL{G} \OTIMES_{\MATHCAL{O}_1} \MATHCAL{O}_2 FLAT OVER \MATHCAL{O}_2

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span's argument is self-contained (displayed change-of-rings identity plus the flatness principle); the author's line writes \mathcal{F} where the test module \mathcal{H} appears on the left, kept verbatim as in the pinned source.

Allowed standard prerequisites (editorial).

- Flatness means the tensor functor is exact. Stacks Project, Modules on Sites, definition of flat module. Supplies that \mathcal{G} \otimes_{\mathcal{O}_1} - is exact, hence so is the isomorphic functor (\mathcal{G} \otimes_{\mathcal{O}_1} \mathcal{O}_2) \otimes_{\mathcal{O}_2} -.

Lemma 0.1. *Let $\mathcal{O}_1 \to \mathcal{O}_2$ be a map of sheaves of rings on a site $\mathcal{C}$. If $\mathcal{G}$ is a flat $\mathcal{O}_1$-module, then $\mathcal{G} \otimes_{\mathcal{O}_1} \mathcal{O}_2$ is a flat $\mathcal{O}_2$-module.*

Proof. This is true because

$$(\mathcal{G} \otimes_{\mathcal{O}_1} \mathcal{O}_2) \otimes_{\mathcal{O}_2} \mathcal{H} = \mathcal{G} \otimes_{\mathcal{O}_1} \mathcal{F}$$

(as sheaves of abelian groups for example). □

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